Kawshik Manikantan

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Education

International Institute of Information Technology (IIITH) Hyderabad, India
B.Tech. (Honours, 2023) + M.S. by Research (2025), Computer Science (Dual Degree) *Jul 2019 – Jan 2025*

- **CGPA:** 9.33/10 (M.S.), 9.26/10 (B.Tech. Honours).
- **Awards:** Dean's List (top 10–20%) for 5 consecutive semesters.
- **Thesis:** *Coreference without Bells and Whistles*; **advisors:** Prof. Vineet Gandhi and Prof. Makarand Tapaswi.
- **Coursework:** Advanced NLP, Computer Vision, Statistical Methods in AI, Advanced Graph Learning, Information Retrieval and Extraction, Digital Image Processing, Data Structures and Algorithms, Operating Systems and Networks, Design and Analysis of Software Systems.
- **Teaching:** Teaching Assistant for Statistical Methods in AI (Fall 2023).

Publications

MINERVA-Cultural: A Benchmark for Cultural and Multilingual Long Video Reasoning [data]
D. Singh, A. Nagrani, ***K. Manikantan****, H. Singh, D. Tewari, T. Weyand, C. Schmid, A. Angelova, S. Dave* ***CVPR 2026 (Highlight)***

IdentifyMe: A Challenging Long-Context Mention Resolution Benchmark for LLMs [code]
K. Manikantan*, M. Tapaswi, V. Gandhi, S. Toshniwal* ***NAACL 2025 (Oral)***

Major Entity Identification: A Generalizable Alternative to Coreference Resolution [code]
K. Manikantan*, S. Toshniwal, M. Tapaswi, V. Gandhi* ***EMNLP 2024***

Research Experience

Pre-Doctoral Researcher | *Python, JAX, Bazel, Mercurial* Jul 2025 – Present
Google DeepMind, Languages Group *Bengaluru, India*

- Developing **agentic training methods** for efficient **deep visual search** over long videos.
- **Post-training Gemini** to improve video reasoning across multilingual and multicultural contexts; co-authored the **MINERVA-Cultural** benchmark (**CVPR 2026, Highlight**).
- **White-box analysis** of cross-lingual factual retrieval in Gemini.

Research Fellow Intern | *PyTorch, Hugging Face Transformers & Datasets* Dec 2024 – Jun 2025
Advanced Micro Devices (AMD), AI Group *Bengaluru, India*

- Improved **continual model-editing** generalization accuracy by ∼**15%** through multi-faceted data augmentation.
- Built a **continually updated, large-scale benchmark** of recent Wikipedia edits to measure how well edited models adapt to new facts.

Undergraduate Researcher | *PyTorch (Lightning, Geometric), NLTK* Apr 2021 – Jan 2025
Center for Visual Information Technology (CVIT), IIITH *Hyderabad, India*

- Proposed **Major Entity Identification (MEI)**, a generalizable alternative to coreference resolution that transfers across domains and enables LLM evaluation (**EMNLP 2024**).
- Built **IdentifyMe**, a long-context mention-resolution benchmark for frontier LLMs (**NAACL 2025, Oral**).
- Re-implemented *Tracking at 100 FPS* (GOTURN) and the *neural MOT solver* (MPNTrack), extending graph-based tracking to **multi-view detection and tracking** [code].

Research Intern | *OpenCV, C++* Jul 2022 – Sep 2022
Star Sports *Remote, India*

- Deployed an **operator-free player-tracking system** used live during the **Asia Cup 2022** broadcasts.
- Drove ideation and feature development for **player segregation**, and built the pre-deployment pipeline.